

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway, EliteUS Software Solutions

Aug 2024 - Present | Piscataway, NJ

- Developed an AI chatbot using **NLP, AWS Bedrock, LLMs, and Streamlit** for system instructions with Python. Leveraged **advanced language models** for seamless deployment and real-time communication.
- Integrated **vector databases** like pinecone, chromadb, **text-image matching**, and **LangChain** for dynamic interactions and enhanced response accuracy through **Retrieval-Augmented Generation (RAG)**.
- Developed and optimized **stored procedures in SQL** using AutoGen to automate complex data transformations, manipulations inturn improving query performance, reducing processing while focusing on quality checks.

Data Science Intern, Harvest Software Solutions, LLC

June 2023 - Sept 2023 | Jacksonville, Florida

- Enhanced real-time ad campaigns by implementing **Machine Learning** models, resulting in a 70% improvement in predictive accuracy and efficiency. Performed **exploratory data analysis** on large-scale datasets.[5M+ rows]
- Collaborated with **cross-functional teams** to translate business requirements to data-driven solutions. Developed interactive dashboards in **Tableau**, effectively communicated strategic insights, business ideas to the client.

Application Development Associate, Accenture Solutions Pvt. Ltd.

June 2021 - Aug 2022 | Hyderabad, India

Database Analyst, Regeneron Pharmaceuticals Inc. - Client Data Analysis & Warehousing

- Optimized **ETL** workflows and implemented **automated data pipelines in Python**, enhancing data processing efficiency.
- Conducted large-scale data analysis using **SQL** and **AWS Athena, Redshift** while visualizing insights and built interactive dashboards in **Tableau** effectively communicating findings to both technical and non-technical stakeholders.
- Increased system reliability by 25% through **DDL optimization, seamless data integration, and performance tuning**, ensuring stable **data warehousing** and reporting operations using **SQL Workbench**.

Research Lab, Andhra University

January 2021 - June 2021 | Visakhapatnam, India

Self Driving Cars Using Image Processing and Deep Learning

- Engineered **Convolutional Neural Networks (CNNs)** for autonomous lane detection and object recognition using python.
- Leveraged **YOLO and Faster R-CNN** architectures to optimize real-time detection and classification accuracy.
- Enhanced model generalization through **image pre-processing** techniques like **Gaussian blurring and color space conversion**, and integrated **multi-sensor data** from **radar, lidar, and cameras** to improve navigational precision.
- Developed a real-time simulation environment using **Unity3D and a Flask backend** for dynamic parameter monitoring and system validation, enabling high-fidelity testing of autonomous driving functionalities.

TECHNICAL SKILLS

Programming: Python, R, Scala, MATLAB. **Libraries:** Pytorch, Scikit-learn, Tensorflow, SparkML, Tableau, PowerBI.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Statistical Analysis: Causal Inference, Hypothesis testing, Time-series forecasting, stochastic process, Linear Regression.
Deep Learning: Neural Networks, Computer Vision, AI, NLP, BERT, GANs, LLMs, LLaMA, LSTM, OpenAI models.
Database & Cloud: SQL, AWS, RDBMS, Data Warehousing, OpenAI, Anthropic, Gemini, GCP, Langchain, Pinecone.
Language Models: OpenAI, Azure, AWS Bedrock, Hugging Face Transformers, LLaMA, prompt tuning, RAG, Langchain.

EDUCATION

Rutgers University, Masters in Data Science

Sept 2022 - May 2024 | New Brunswick, NJ

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** [Portfolio](#).

Andhra University, B.Tech in Electrical and Electronics Engineering

2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Twitter Search Application - [PostgreSQL, MongoDB, Python, Sorting techniques, Search Engine Optimization]

Created a user query-responsive tweet retrieval system by integrating dual databases: Postgres for user data management and MongoDB for tweet storage. Optimized data storage and retrieval efficiency while implementing advanced caching strategies.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Deaths caused due to drug overdose - [R programming, Geospatial Analysis, Temporal Analysis, plotly, ggplot2, Data Analysis] This research investigates the drivers and trends of drug abuse in Connecticut, using a data-driven approach to analyze demographics, temporal patterns utilising R programming. The goal is to provide a comprehensive understanding that informs effective interventions and policies.

Thank you for considering my application. I'm available to start working immediately.